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System-Prompt Anchoring with Cross-Attention Layers

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TL;DR — Cross-Attention Layer (CAL) blocks give a frozen causal decoder a dedicated route to the privileged system-prompt span. A ten-configuration sweep at 1.5B shows that *where* the route is inserted matters: later placements are generally more effective and parameter-efficient. At 8B, the selected LATE8TH configuration improves instruction adherence and many-shot jailbreak resistance over a parameter-matched baseline, with task-dependent effects elsewhere.

sole author · Qwen2.5-1.5B · Llama-3.1-8B · 11 benchmarks · arXiv 2605.09737

MOTIVATION

System rules and untrusted text get equal priority

1 INSTRUCTION HIERARCHY

- Deployed LLMs rely on a **system prompt** for behavioral constraints and safety rules.
- Standard causal self-attention treats privileged instructions and untrusted user data with **equal priority**.
- **Prompt injection:** adversarial user inputs overwrite the system prompt's rules and hijack the model's objective.
- **Context drift:** system instructions gradually erode over extended or multi-turn sequences.

2 RELATED APPROACHES

- **Behavioral alignment:** SecAlign and StruQ use preference optimization and structured queries; Instruction Hierarchy training teaches models to prioritize privileged instructions.
- **Representation-level:** AIR injects hierarchy signals into intermediate representations; ASIDE separates instructions and data at the embedding level.
- **Cross-attention routes:** encoder–decoders, RETRO, Flamingo and BLIP-2 condition on a separate source; ControlNet uses zero-initialized branches on a frozen backbone.

3 THIS PAPER

CAL blocks source keys and values from a privileged span *within the same causal sequence*. The focus is not a new cross-attention architecture, but how injection depth affects use of that route.

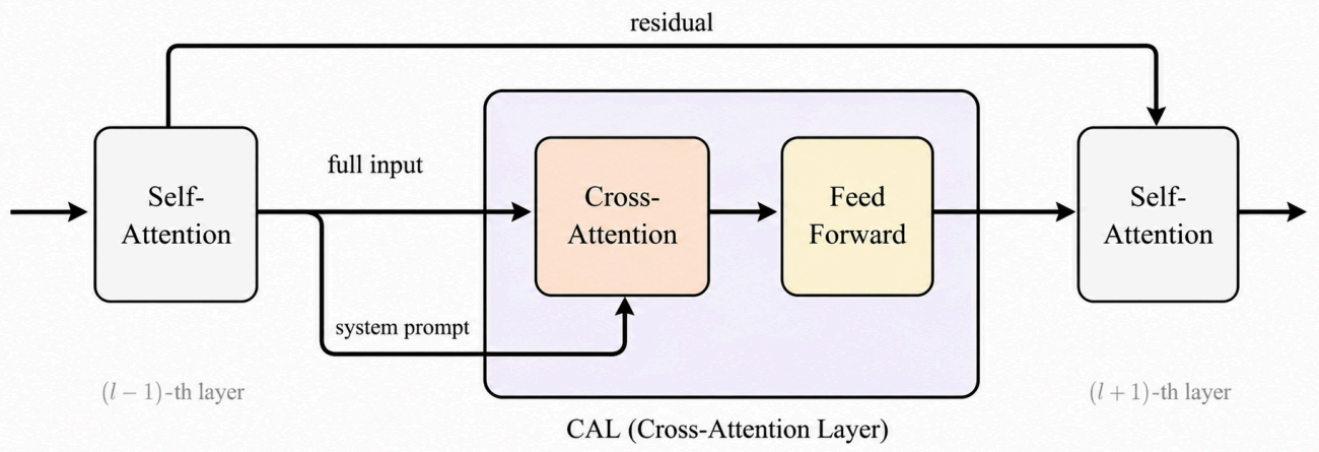
Q1. Does a dedicated route to the system prompt change performance on security and instruction-following tasks?

Q2. How does the placement of the injected blocks affect that performance?

METHOD

A CAL block: cross-attention into the system-prompt span

1 ARCHITECTURE



Blocks are inserted after selected decoder layers. Each lets the full residual stream query the system-prompt span, then applies a SwiGLU feed-forward sublayer; the ordinary self-attention and residual paths stay intact.

- Backbone **frozen**; only CAL parameters are trained.
- Output projections W_O, W_D are **zero-initialized**, so every inserted block is an identity map at initialization.

2 CAUSAL CROSS-ATTENTION

Queries come from the full sequence; keys and values only from the system span $[s, e)$, after RMSNorm:

$$\mathbf{Q} = \overline{\mathbf{X}} W_{\mathrm{Q}},\ \mathbf{K} = \overline{\mathbf{X}}_{[s:e]} W_{\mathrm{K}},\ \mathbf{V} = \overline{\mathbf{X}}_{[s:e]} W_{\mathrm{V}}$$

A query at position i may attend to span slot j only if $j \le i - s$: system tokens see only their causal prefix, while later user and generated tokens see the whole span, with no future-token leakage.

$$\mathbf{X}' = \mathbf{X} + \mathbf{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}, \mathbf{M})\, W_{\mathrm{O}}$$
$$\mathbf{X}'' = \mathbf{X}' + \mathbf{SwiGLU}(\mathbf{RMSNorm}(\mathbf{X}'))\, W_{\mathrm{D}}$$

Span bounds and CAL keys/values are computed at prefill and cached for decoding.

EXPERIMENTAL SETUP

Ten placements, two matched controls, eleven benchmarks

1 PLACEMENTS ON 28 LAYERS

CONFIG	PLACEMENT	BLOCKS
ALL	After every layer	28
EVERY2	After every other layer	14
EARLYTHIRD	Layers 1–11, contiguous	11
MIDTHIRD	Layers 9–19, contiguous	11
LATETHIRD	Layers 18–28, contiguous	11
LATEHALF	Last ½, contiguous	15
LATEHALF2	Last ½, interleaved	8
LATEQUARTER	Last ¼, distributed	8
LATE8TH	**Last ⅛, distributed**	**5**
LAST	After the final layer	1

2 MATCHED CONTROLS

Both match the trainable-parameter count of LATE8TH and use the same data.

- **LoRA:** standard full-network adaptation (rank 164 at 1.5B, 512 at 8B).
- **ParallelMLP:** zero-initialized MLP adapters parallel to the same blocks as LATE8TH, on a frozen backbone. Controls capacity, location and freezing, but not information routing.

Trainable parameters at 1.5B: CAL 188,759,040 · LoRA 189,263,872 · ParallelMLP 188,751,360.

3 DATA AND EVALUATION

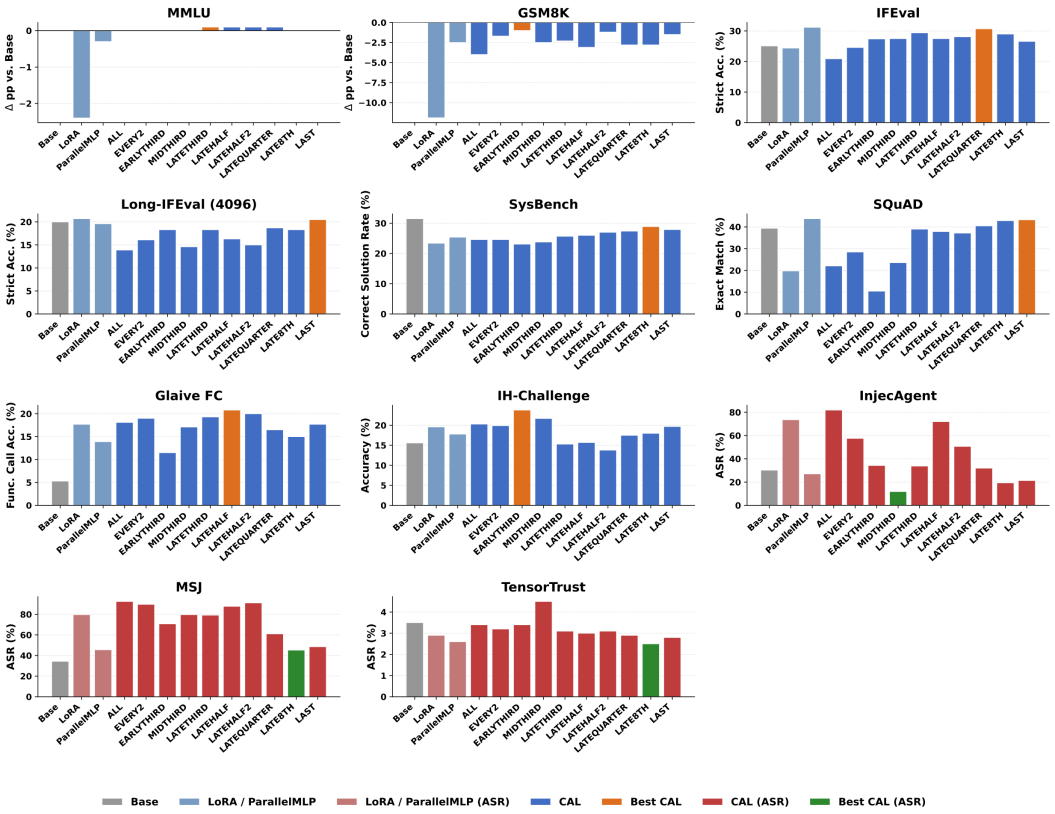
50,000-example general SFT corpus: 40% OpenHermes / 40% Airoboros / 20% PKU-SafeRLHF.

GENERAL
MMLU, GSM8K

INSTRUCTION FOLLOWING
IFEval, Long-IFEval (4K–32K tokens), SysBench, SQuAD, Glaive FC

SECURITY
IH-Challenge, InjecAgent, Many-Shot Jailbreaking (MSJ), TensorTrust

Placement matters, and later is generally better



MMLU and GSM8K: pp change vs. Base; other panels absolute. Blue = accuracy (higher is better); red = attack success rate (lower is better).

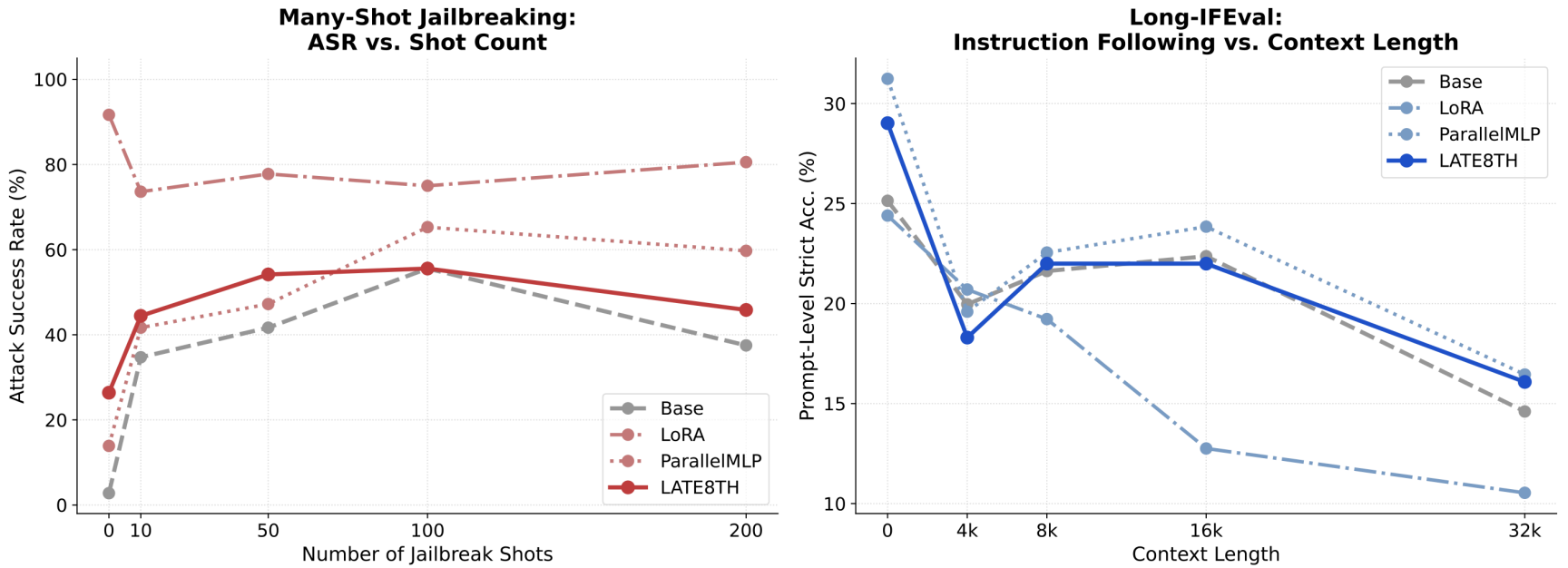
1 DEPTH EFFECT

- **Sparse late placements** generally offer the strongest performance–parameter trade-off; early or dense insertion performs poorly on several instruction-following and security tasks.
- **No placement dominates:** different depths lead IFEval, Glaive FC, IH-Challenge and InjecAgent.
- Within late placements, distributing a few blocks beats concentrating the intervention in one final block.

2 MATCHED CONTROLS

- **LoRA** loses substantial MMLU and GSM8K performance and sharply increases attack success on InjecAgent and MSJ.
- **LATE8TH vs. ParallelMLP:** comparable on formatting and extraction, stronger on SysBench and Glaive FC, narrowly better on all four security evaluations.

More jailbreak shots, longer contexts



LORA

Degrades rapidly outside its training distribution, which contains neither extreme-length nor adversarial multi-turn examples.

MANY-SHOT JAILBREAKING

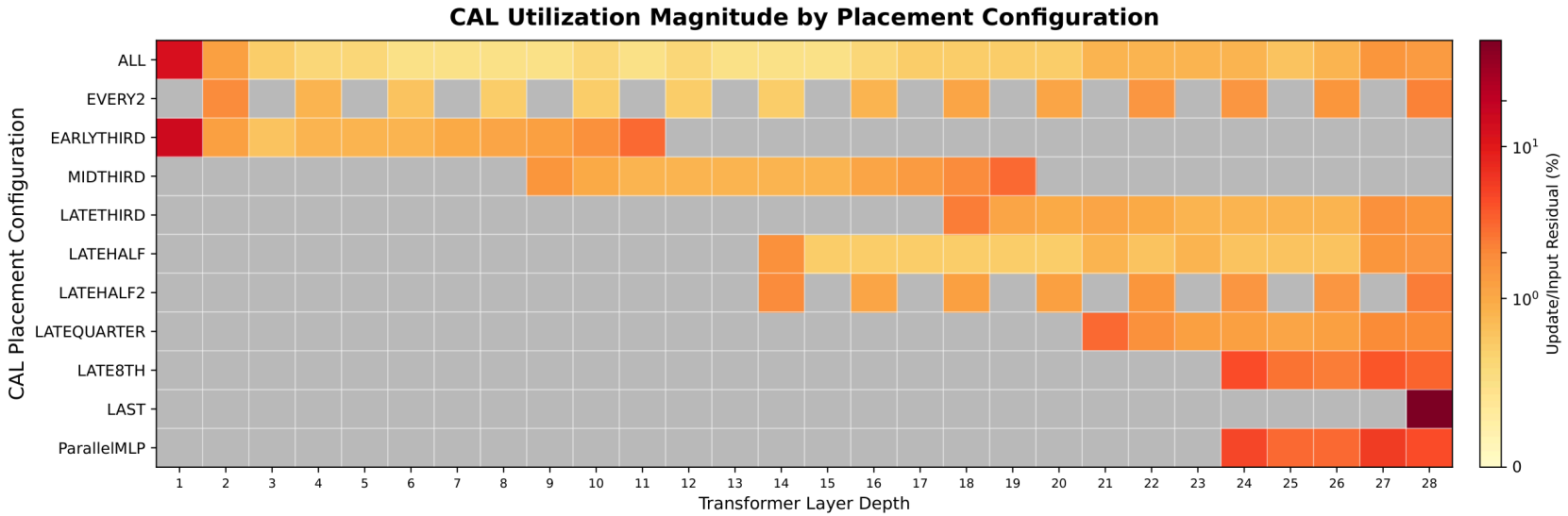
At 200-shot MSJ, LATE8TH outperforms ParallelMLP (lower attack success rate is better).

LONG-IFEVAL

LATE8TH degrades smoothly at extreme lengths but does not surpass ParallelMLP in absolute accuracy.

MECHANISTIC ANALYSIS

How much does each inserted block update the residual stream?



Cell = $100 \cdot E\|X'' - X\|_2 / E\|X\|_2$ for one configuration and layer; gray = no inserted block.

WITHIN-WINDOW PATTERN

Updates spike at the first available block, fall through the middle, and rise toward the end, for EARLYTHIRD and MIDTHIRD alike.

LATE CONFIGURATIONS

Strong late configurations keep updates above 1% near the end of the network; ParallelMLP has a similar late-stage profile to LATE8TH.

NOT A PERFORMANCE PROXY

Dense early placements and single-block LAST make large updates without being the strongest models.

RESULTS · TRANSFER TO LLAMA-3.1-8B-INSTRUCT

At 8B, late8th separates from its matched control

	General		Instruction following						Security		
Model	MMLU ↑	GSM8K ↑	IFEval ↑	L-IFEval ↑	SysBench ↑	SQuAD ↑	Glaive ↑	IH-Chal. ↑	InjecAgent ↓	MSJ ↓	TensorTrust ↓
Base	68.25	85.37	71.16	41.40	60.76	47.50	14.78	11.13	82.00	10.00	7.83
LoRA	64.88	76.95	57.12	34.66	45.82	64.55	21.87	20.79	84.75	73.61	5.05
ParallelMLP	68.01	83.02	61.74	31.84	42.32	59.33	1.89	12.01	71.55**	38.05	7.22
CAL (late8th)	**68.25**	**83.02**	**69.13***	**33.92**	**58.60****	**58.75**	**6.66****	**14.14****	**81.90**	**25.00****	**6.80**

All values %; ↓ = attack success rate (lower is better). Stars mark a significant ParallelMLP vs. CAL difference: ** p<0.01, * p<0.05 (two-sided proportion z-test).

+7.4 pp
IFEval vs. ParallelMLP (69.13 vs. 61.74)

+16.3 pp
SysBench vs. ParallelMLP (58.60 vs. 42.32)

38.05 → 25.00%
Many-shot jailbreak ASR, ParallelMLP → CAL

NOT UNIFORM
SQuAD nearly unchanged, TensorTrust only marginal, and InjecAgent favors ParallelMLP. Both frozen adapters preserve general ability far better than LoRA.

TAKEAWAYS

Placement is a design variable, not a detail

1 CONCLUSIONS

- Placement is a major, **task-dependent** factor; later blocks are generally strong and parameter-efficient.
- At 8B, the transferred LATE8TH separates from matched ParallelMLP on IFEval, SysBench and many-shot jailbreaking.
- The route helps most with explicit constraints, multi-turn adherence and sustained many-shot attacks, not every form of extraction or injection.

→ FUTURE WORK

- Other tasks, including privileged sources that are not system prompts.
- Repeat sweeps across corpora, model families, scales and seeds; adversarial and preference-based training of the inserted blocks.

2 LIMITATIONS

- One frozen-backbone setup and one 50,000-example corpus: rankings may depend on corpus, optimization recipe, backbone family and freezing.
- Each configuration was trained once, so variation across runs and initializations is not measured.
- Placements swept only on Qwen2.5-1.5B. The 8B study changes model family, tokenizer, chat template and hyperparameters together, so it cannot isolate scale.
- CAL does not uniformly beat the Base model on security, and is not a hard security boundary: it should not replace access control or input sanitization.

3 COST AT 1.5B (5 BLOCKS)

+12.23%
parameters (188.76M), +360 MB bf16 weights

+18.2–19.9%
median prefill overhead, 512–4096-token prompts

+12.0–13.2%
decode overhead (time-to-first-token: +17.9–19.9%)

Measured at batch size one with bf16 FlashAttention-2 on an RTX 4080 Laptop GPU; the stable percentage overhead is consistent with caching CAL keys/values.